

Tutorials for the GRASS geocomputation engine

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Software

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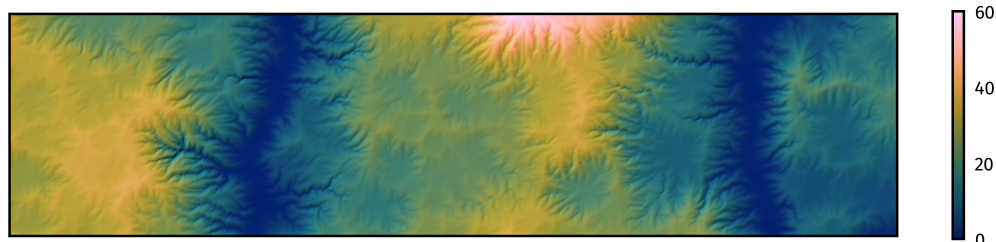


Figure 1: An example of fractal terrain generated with GRASS and eroded with LandLab from a tutorial on scientific modeling. This tutorial demonstrates a seamless workflow for scientific modeling in Python, showing how gridded data can be passed as arrays between GRASS, NumPy (Harris et al., 2020), and Landlab (Barnhart et al., 2020).

Summary

This collection of tutorials is an introduction to the GRASS geospatial processing engine. GRASS is an open source computational engine for spatiotemporal data management, analysis, modeling, and simulation (GRASS Development Team et al., 2025; Neteler & Mitasova, 2008). As an engine that can be integrated in data science pipelines with shell scripting, Python, R, Jupyter, and Colab, there are many ways to use GRASS. While GRASS already had extensive documentation of individual processing tools, tutorials were needed to introduce the many ways to interface with the tools and combine them into computational workflows (Figure 1). These open education tutorials - which cover integrations, core features, and disciplinary applications - were developed as part of an effort to grow the GRASS community. The tutorials are built with Quarto and are deployed as webpages paired, when appropriate, with Jupyter computational notebooks. The tutorials are available at <https://grass-tutorials.osgeo.org> under both the GNU Free Documentation License v1.2 or later and the Creative Commons Attribution-ShareAlike 4.0 International License.

Statement of Need

As GRASS has grown from its roots as a geographic information system (Westervelt, 2004), it has evolved into a geocomputational engine with many interfaces. As an engine, it can be integrated in geospatial data science pipelines using shell scripting, application programming interfaces, tangible interfaces (Petrasova et al., 2018), computational notebooks (Haedrich et al., 2023), cloud computing environments (Neteler et al., 2019; White et al., 2023), or high performance computing environments. While GRASS is well documented with books, a user manual, developer manuals, and a wiki, it lacked

29 official tutorials. Over the years, the community developed many tutorials across different
30 platforms, but as these are independently maintained, many have become outdated and
31 obsolete. The current roadmap for GRASS – established in 2024 – calls for official new
32 tutorials to encourage community growth and demonstrate integrations in data science
33 pipelines.

34 The design and implementation of the new official tutorials for GRASS was based on
35 experience teaching university courses and conference workshops using open educational
36 resources. Over the last decade, the GRASS community has developed many open
37 educational resources, experimenting with delivery via web documents, computational
38 notebooks, and cloud computing services. Online tutorials for GRASS have been built
39 from source in HTML (Petrás et al., 2015), built from Markdown with a static site
40 generator (Harmon, 2020), included Jupyter notebooks (Haedrich et al., 2023), and used
41 cloud computing services such as Binder (Petrásova & Petráš, 2019), Whole Tale (Andreo,
42 2023b), and Google Colab (Andreo, 2023a). Petráš et al. (2015) used a modular structure
43 with tabsets to teach the core interfaces for GRASS – the GUI, CLI, and Python API –
44 separating explanatory text introducing geospatial concepts from software specific text
45 for each interface. This scaffolding helps learners to focus on concepts, while building
46 their skills with increasingly complex interfaces. Haedrich et al. (2023) developed the
47 GRASS–Jupyter integration to incorporate more scripting into a graduate-level course on
48 geospatial computing and simulation. The package extends the existing GRASS Python
49 APIs with data visualization and management tools for the Jupyter environment. The
50 new course materials include Jupyter Notebooks that combine tutorials and assignments,
51 allowing students to write and modify code, interact with examples, and explain their
52 reasoning in markdown, all within a single document. Based on these experiences, our
53 design principles for the new tutorials include teaching geospatial concepts discretely
54 from software specifics to encourage spatial thinking, supporting live coding to encourage
55 computational thinking, and using an open source publishing system to build documents
56 from plain text tracked with version control.

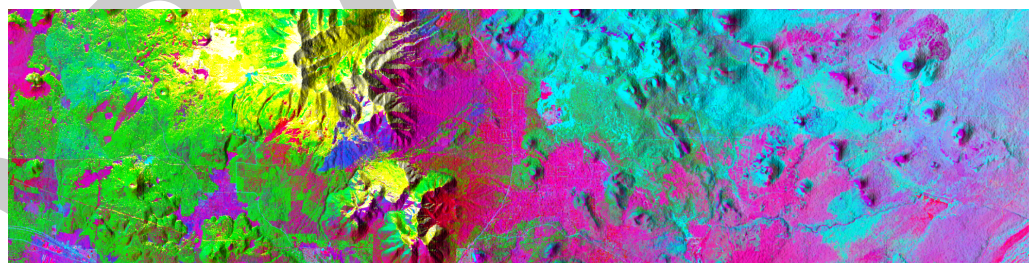


Figure 2: An example of image fusion of principal components analysis of multi-band images of the San Francisco volcanic field from a tutorial introducing the basics of remote sensing in GRASS. This tutorial demonstrates how to process and visualize multi-band remote sensing imagery.

57 Description

58 Learning Objectives

59 This collection of tutorials was designed to teach geocomputational thinking using the
60 GRASS geoprocessing engine. To introduce computational approaches to thinking
61 (National Research Council, 2010; Weintrop et al., 2016) about space and time, the
62 tutorials cover the fundamentals of geoprocessing with GRASS, integrations of GRASS
63 into data science pipelines, and disciplinary applications of GRASS. The tutorials were
64 designed for self-study by learners of all levels, integration into courses, and deployment
65 in workshops.

Instructional Design

In order to teach a computational approach to thinking about spatiotemporal phenomena through different interfaces to the GRASS engine, the tutorials were designed:

- as modules for reuse and remixing,
- as worked examples to reduce cognitive load,
- as interactive lessons for active learning and engagement,
- as scaffolded prose and code to structure learning,
- and as computable content to teach computational thinking.

Drawing on the education benefits of computational notebooks (Barba et al., 2022), the tutorials introduce geocomputational concepts through worked examples that synthesize prose explanations, graphics, and executable code. The tutorials, which range from introductory to advanced, have a modular design for reuse and remixing so that learners can choose their own course of study and teachers can select modules for their lesson plans. The tutorials build in complexity from introductory to core to disciplinary modules. A set of getting-started tutorials introduce different ways to interface with the GRASS engine. The core tutorials cover important concepts such as geovisualization (Figure 2), map algebra (Figure 3), geospatial modeling, and the temporal framework. Disciplinary tutorials demonstrate applications for GRASS in domains such as climatology, ecology, hydrology, geomorphology, and terrain generation (Figure 4). The disciplinary tutorials build engagement by working through applications in the learner's domain and thus motivate further exploration. Throughout the tutorials, different ways to interface with GRASS are presented as tabs in code blocks, so that learners can work their way through the same tutorial repeatedly using increasingly challenging interfaces – building proficiency first with the graphical user interface (GUI), then with the command line interface (CLI), and finally with the Python or R application programming interfaces (API).

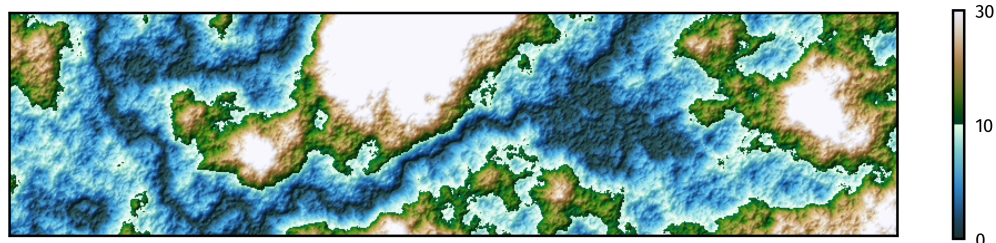


Figure 3: An example of synthetic terrain generated using map algebra from a tutorial introducing the basics of map algebra in GRASS. This tutorial demonstrates local algebraic operations using the raster map calculator, focal operations using nearest neighbors analysis, zonal operations using conditional statements with the raster map calculator, and global operations using raster metadata.

Implementation

To reach a broad audience, the tutorials in this collection are published as web documents for immediate, easy access via web browsers. When appropriate, web documents are accompanied by a downloadable computational notebook, encouraging interactivity, engagement, and geocomputational thinking. The tutorials – which are built and deployed using the Quarto scientific publishing system (Allaire et al., 2025) – are written in Markdown with YAML frontmatter. Tutorials are composed in Markdown for human-readable source code, efficient version control, executable code blocks for different interfaces, rendering in multiple formats, and reproducibility. As this open education project aims to teach different ways to interface with GRASS, executable code for multiple relevant interfaces such as the GUI, CLI, Python, or R can be included in tutorials as tabsets.

102 Once tutorials have been written, they are reviewed by the GRASS Development Team,
103 rendered as web documents and Jupyter notebooks, and deployed to an Open Source
104 Geospatial Foundation website. The source code for the tutorials is built in the GitHub
105 [repository](#) and deployed to the website <https://grass-tutorials.osgeo.org> using GitHub
106 Actions.

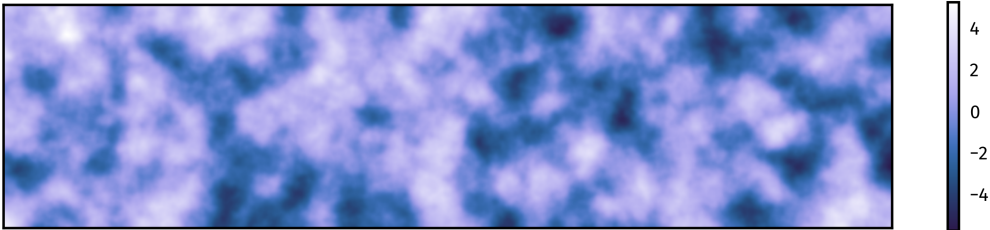


Figure 4: An example of fractional Brownian motion from a [tutorial](#) introducing procedural noise in GRASS. This tutorial demonstrates stochastic functions for procedurally generating data.

107 **Content**

108 This official collection of tutorials is maintained by the GRASS Development Team as part
109 of the documentation for the GRASS geocomputational engine ([Table 1](#)). This ensures
110 that tutorials are standardized, undergo rigorous review, and are maintained and updated
111 as GRASS evolves. The website also includes a curated collection of external tutorials
112 that are hosted and maintained by their creators ([Table 2](#)).

Table 1: Official GRASS tutorials

Modules	Tutorials	Level	Lang.
Integrations	Get started with GRASS GUI	Beg	En
	Get started with GRASS & Python in Jupyter Notebooks	Beg	En
Core	Get started with GRASS in Google Colab	Beg	En
	Getting started with GRASS for Conda	Beg	En
	Get started with GRASS in Jupyter Notebooks on Windows	Beg	En
	Quick comparison: R and Python GRASS interfaces	Int	En
	Get started with GRASS & R: the rgrass package	Adv	En
	Basics of map algebra	Beg	En
	Making plots with GRASS	Beg	En
	Visualizing and modeling terrain from DEMs	Beg	En & Pt
	Introduction to remote sensing with GRASS	Beg	En & Pt
	Making thematic maps	Beg	En
	Introduction to time series in GRASS	Int	En
	Temporal subset, import and export	Int	En
	Temporal aggregations	Adv	En
	Temporal algebra	Adv	En
	Temporal accumulation	Adv	En
	Temporal gap-filling	Adv	En
	Temporal query with vector data	Adv	En
Disciplinary	Modeling movement in GRASS	Adv	En & Pt
	Basic earthworks	Beg	En
	Gully modeling	Beg	En
	Coastal infrastructure	Beg	En

Modules	Tutorials	Level	Lang.
	Terrain synthesis	Int	En
	Procedural noise	Int	En
	Hydro-flattening a digital elevation model	Int	En
	Using GRASS, NumPy, and Landlab for scientific modeling	Int	En
	fasterRaster: faster raster processing in R using GRASS	Int	En
	Estimating wind fetch	Adv	En
	Parallelization of overland flow simulation	Adv	En

Table 2: External tutorials

Modules	Tutorials	No.	Level	Lang.
Integrations	Unleash the power of GRASS	5	Beg-Adv	En
	GRASS for remote sensing data processing with Jupyter Notebooks	1	Adv	En
Core	NCSU geospatial modeling and analysis course	13	Beg-Int	En
	Geoprocessamento com GRASS	1	Beg-Int	Pt
	Tutoriales de GRASS en grasswiki	4	Beg-Int	Es
	GISMentors	30	Beg-Adv	En & Cs
Disciplinary	Deforestation study using GRASS	1	Beg	En
	Teledetección, OBIA y series de tiempo	5	Beg-Adv	Es
	GIS for designers	12	Beg-Int	En
	GRASS for environmental monitoring and disease ecology	2	Beg-Int	En
	Processing lidar and UAV point clouds	1	Beg-Int	En
	Physically-based hydrologic modeling using GRASS r.topmodel	1	Int	En
	Spatio-temporal data handling and visualization	1	Int	En
	Ecodiv.earth tutorials	16	Beg-Adv	En
	Urban growth modeling with FUTURES	1	Adv	En

Acknowledgements

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